

- CareerOrbit V3.1 — Full-Loop QA Test Script
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CareerOrbit V3.1 — Full-Loop QA Test Script

Purpose: A step-by-step human verification protocol to ensure every cognitive and technical system is functioning correctly in a production-like environment.

🏁 Phase 1: Onboarding & Discovery

1. **Action:** Navigate to </onboarding>.
2. **Verify:** Multi-step form renders with glassmorphic animations.
3. **Action:** Complete all 8 steps with valid data.
4. **Verify:** Redirection to </path-selection> occurs within 3 seconds.
5. **Verify:** Three distinct career paths are generated by the AI with ₹ earnings clearly visible.

🧪 Phase 2: The Diagnostic Wall (Pre-Test)

1. **Action:** Select a path and click "Begin Pre-Test."
2. **Verify:** Exactly 10 questions are rendered covering the path's core Knowledge Components.
3. **Action:** Answer 7/10 correctly.
4. **Verify:** REDIRECT to </dashboard>.

5. **Database Audit:** Check the `user_model` table. Verify `pMastery` values are initialized between 0.40 and 0.70.
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Phase 3: The Adaptive Learning Loop

1. **Action:** Open the first module and watch the instructional video.
 2. **Action:** Attempt the "Check Mastery" quiz.
 3. **Verify:** Correct answer triggers a "Success" animation.
 4. **Verify:** $P(L)$ probability in the Dashboard (Radar Chart) increases.
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Phase 4: Proactive Mentor Intervention

1. **Action:** Open a new task and stay idle for 15 minutes OR intentionally fail the quiz twice.
 2. **Verify:** The **AI Mentor** floating button starts pulsing (`animate-bounce`).
 3. **Action:** Open the chat.
 4. **Verify:** The greeting acknowledges the struggle (e.g., "I noticed you're stuck...").
 5. **Action:** Ask a non-vocational question (e.g., "Tell me a joke about politics").
 6. **Verify:** The AI uses the **Refusal Protocol** and redirects back to the task.
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Phase 5: Dynamic Recalibration

1. **Action:** Intentionally fail 3 consecutive quizzes on a Medium-difficulty subtopic.
 2. **Verify:** The system triggers a "Recalibrating..." loading state.
 3. **Verify:** The subtopic is "rewritten" with simpler language and a new video.
 4. **Verify:** $P(L)$ state reflects a regression but remains clamped at 0.05.
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Phase 6: Mastery & Telemetry

1. **Action:** Achieve 100% mastery on a module.
 2. **Verify:** The "Mastery Orb" glows at maximum intensity.
 3. **Verify:** The **Hake Gain** metric in the stats panel is calculated correctly.
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Regression Checklist

- ☐ **Mobile Responsiveness:** Verify the `/architecture` mind-map is navigable on touch devices.
- ☐ **Language Integrity:** Switch to Hindi/Kannada and verify all AI content translates correctly.
- ☐ **Data Isolation:** Log in as User B; ensure User A's roadmap is invisible.